# PHARMACEUTICAL QUALITY ASSURANCE

## UNIT 4 NOTES

### COMPLAINTS

- EVALUATION
- HANDLING OF RETURN GOOD
- RECALLING & WASTE DISPOSAL

### DOCUMENT MAINTAINANCE

- BATCH FORMULA RECORD
- MASTER FORMULA RECORD
- SOP
- QUALITY REVIEW
- REPORTS & DOCUMENTS
- DISTRIBUTION RECORDS

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# COMPLAINTS

- A complaints are any written, verbal statement that reports a problem related to the quality, safety or performance of the product.
- It shows costomer dissatisfaction about the company & the product.

## TYPES OF COMPLAINTS

- Complaints can be following types:

### ① Quality Complaints

- These are complaints related to product defects such as broken tablets, discoloration, leakage.

### ② Adverse Reaction Complaints

- These are related with unexpected side effects or harmful reactions of product.

### ③ Other Complaints

- They include complaints such as lack of efficacy, labelling, container defects etc.

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## Source Of Complaints

In pharma sector, complaints can be raised from following sources :

- Patients or Consumers
- Doctors or Pharmacists
- Hospitals or Distributors
- Regulatory Authorities

## Classification Of Complaints

Based on severity & impact on patients, complaints can be classified as follows :

① Type A (Critical)
② Type B (Major)
③ Type C (Minor)

### ① Type A (Critical)

- These are very serious complaints that may harm the patient.
- They involve safety risks or life threatening situations.
- They require Immediate investigation & possible product recall.

Examples :

- Wrong product or wrong label
- Contamination of product
- Presence of foreign particles (glass, metal)
- Serious adverse drug reactions

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## ② Type B (Major)

- These complaints may affect product quality, but are not life-threatening.
- They can reduce the effectiveness or acceptability of the product.
- They require CAPA (Corrective & Preventive Action)

Examples:

- Broken Tablets
- Improper sealing or leakage
- Variation in colour or appearance

## ③ Type C (Minor)

- These are less serious complaints with no impact on safety or efficacy.
- They're usually related to acceptance or minor defects.
- They should be reviewed, but no urgent action needed.

Examples:

- Slight Packaging defects
- Minor printing issues

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## Handling & Evaluation Of Complaints

- It is the Systematic process of recieving, investigating & analyzing complaints to take CAPA.
- It involves following steps:

① Recieving complaints
② Technical Investigation
③ Corrective & Preventive Action
④ Feedback to Customer
⑤ Monthly Report & Trend Analysis

### ① Recieving Complaints

- Complaints can be recieved from customers, hospitals, distributors etc.
- Company must provide a support system through a toll free number or chat support.
- All complaints must be properly recorded in a product complaint Data sheet.
- Each complaint is given a unique complaint number for tracking.

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### Contents of a product Complaint Data Sheet

- Serial number assigned to the complaints
- Exact nature of the complaints
- Name & Address of the complainant.
- Date of the complaint recieved
- Name of the product, strength & batch number of the product
- Quantity involved in the complaint
- Size of the sample obtained by QC department
- Name & signature of the investigator & date
- Action taken by the company
- Copy of reply sent to complainant.

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## ② Technical Investigation

- It is the most important step in complaint handling.
- In this step, we scientifically analyze the complaint to find the root cause & verify whether the complaint is valid or not.
- It involves two steps:
  - Documentation - Based Investigation
  - Laboratory Analysis

### a) Documented Based Investigation

- We review all related records, such as:
  - Batch Manufacturing Record (BMR)
  - Batch Packaging Record (BPR)
  - Quality Control (QC) data
- We check any error during manufacturing or packaging.
- This helps to identify if the problem occured during production.

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### b) Laboratory Analyse Phase

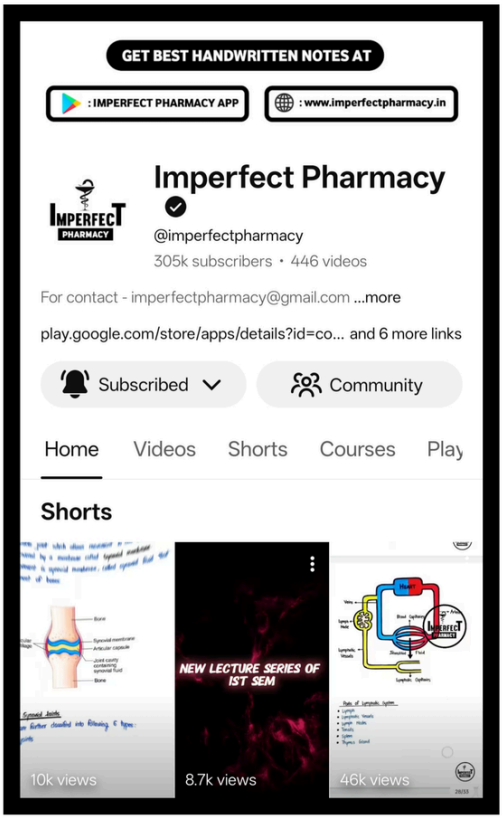
- We test the complaint sample & retained sample in the laboratory.
- This confirms whether the defect is present in the complaint sample or whole batch.

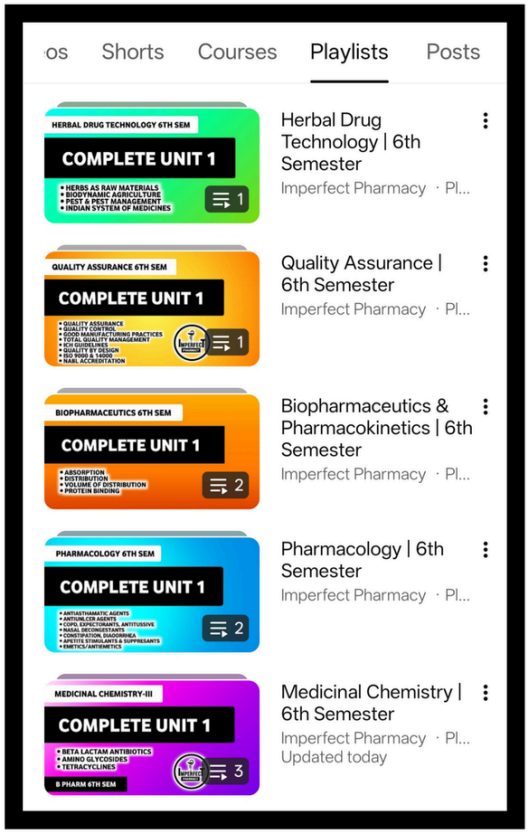
### Evaluation Of Results

- After investigation, we classify the complaint as:
  - Confirmed complaint: Defect is found, Complaint is valid.
  - Non-Confirmed complaint: No defect found, Complaint is invalid.

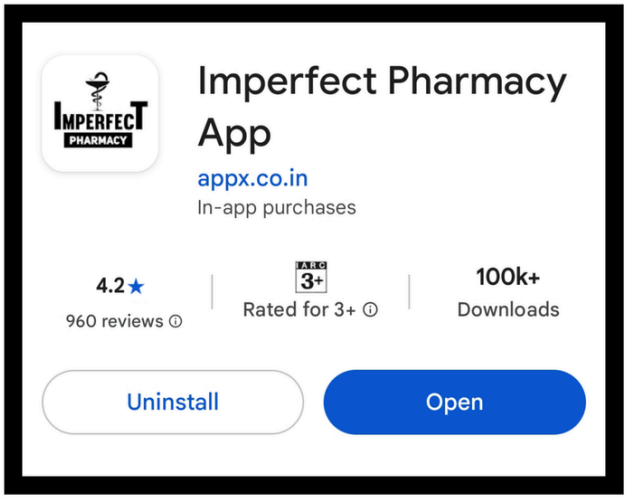
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### Special Case : Counterfiet / Tampering Suspicion

- IF product look suspicious, packaging, labelling & batch detail are verified & information is sent to regulatory authority to take strict action.
- This ensures protection against fake or tampered product.

## ③ Corrective & Preventive Control

- After completing the investigation, the company takes actions based on the root cause.
- Corrective Action means fixing the current problem.
  - Example: If tablets are breaking, adjust compression force.
- Preventive Action means stopping the problem from happening again in future.
  - Example: Improving process control or training staff.
- CAPA ensures continuous improvement & quality assurance.

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## ④ Feedback To Customer

- After investigation & CAPA, the company provides a response to the customer.
- The feeback includes:
  - Result of investigation
  - Explanation of the issue
  - Actions taken
- Communication should be clear & professional.
- This step helps in building customer trust & satisfaction.

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## ⑤ Monthly Report & Trend Analysis

- All complaints are recorded & reviewed regularly.
- A monthly report is prepared following:
  - Number of complaints
  - Type of complaints
  - Batch-wise details
- Trend analysis is done to:
  - Identify repeated or common problems
  - Detect increase in specific defects
- If any trend is observed immediate action is taken & process improvement is implemented
- This helps in early detection of issues & long-term quality improvement.

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## Time Period For Investigation Of Complaints

- Product quality Complaint : 10 Days
- Adverse Reaction Complaint : 5 Days
- Medical Complaint : 3 Days

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## Records & Maintinance

- Record & Maintenance of complaints means proper documentation, storage & review of all complaints recieved for p'ceutical products.
- All complaints should be recorded in a complaint register or data sheet.
- Each complaint must have a unique identification number.
- All investigation details & CAPA actions must be documented.
- Records should be properly maintained & easily retrievable.
- Complaint records must be reviewed periodically for trend analysis.
- All records should be retained for a specified time as per regulatory requirements.

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# Handling Of Return Good

- Returned goods are p'ceutical products that are sent back from the market (customers, distributors or retailers) to the company due to reasons like damage, expiry, defect or excess stock.

## Reason For Return

- Damaging during transportation
- Expiry or near expiry
- Products defects or complaints
- Incorrect supply of product
- Excess stock in the market

## Steps In Handling

### ① Reciept Of Returned Goods

- Returned products are recieved & recorded properly.
- Details like product name, batch number & reason for return are noted.

### ② Inspection & Verification

- The product is checked for:
  - Physical condition (damage, leakage)
  - Seal integrity
  - Labeling & packaging
- It is verified whether the product is genuine & not tampered.

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## ③ Evaluation

- Quality team evaluates whether the product is :
  - Safe for reuse
  - Needs reprocessing
  - Should be rejected

## ④ Final Decision

- Based on evaluation, the product is :
  - Re-stocked : if product is in good condition
  - Reprocessed / Repacked : If minor issues are present.
  - Destroyed : If product is damaged, expired, or unsafe.

## ⑤ Documentation

- All documents are properly recorded.
- Records must be maintained for future reference & audit.

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# DRUG RECALL

- Drug Recall is the process of removing or withdrawing a defective or potentially harmful product from the market by the manufacturer or regulatory authority to protect public health.
- It doesn't include the normal removal of products that have passed their expiry period.

## TYPES OF RECALL

Product Recall can be of two types:

1. Voluntary Recall
2. Statutory Recall

## 1. VOLUNTARY RECALL

- The company itself initiates the recall of the product.
- It is done when the company identifies a defects or safety issue.

## 2. STATUTORY RECALL

- The recall is ordered by regulatory authority (CDSCO)
- It is compulsory & legally enforced.
- Done when the product violates safety or quality standards.

## REASON FOR DRUG RECALL

- Quality defects (e.g., contamination, wrong composition)
- Adverse effects or safety issues
- Labeling or packaging errors
- Regulatory non-compliance

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# Classes Of Drug Recall

## ① Class I (Serious Risk)

- These are products that may cause serious health problems or death.
- They require immediate recall.

## ② Class II (Moderate Risk)

- Products that may cause a temporary or medically reversible health effects with low risk of serious harm.
- Recall is important but not extremely urgent.

## ③ Class III (Minor Defects)

- Products that are not likely to cause any health risk, usually related to minor defects like labelling or packaging errors.
- Doesn't cause significant health effect.

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## Steps In Drug Recall

- Identification of defective product
- Decision to recall
- Inform regulatory authority
- Notify distributors & retailers
- Withdraw product from market
- Documentation & final report

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# Waste Disposal

Waste disposal is the process of collecting, treating & safely disposing of waste materials generated during p'ceutical manufacturing to protect health & the environment.

## Types Of Pharmaceutical Waste

1. Solid Waste: Packeging materials, paper, plastics
2. Liquid Waste: Chemical solutions, Solvents
3. Biological Waste: Cultures, Contaminated materials
4. Hazardous Waste: Toxic, flammable, or reactive substances

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## Steps In Waste Disposal

### 1. Separation Of Waste

- Waste is separated based on type (solid, liquid, hazardous).
- Different colour-coded containers are used.
- Prevents mixing of hazardous & non-hazardous waste.

### 2. Collection & Storage

- Waste is collected in labelled containers.
- Stored in designated areas under controlled conditions.
- This avoids leakage & contamination.

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## ③ Disposal

- Waste is disposed using following methods:
  - Incineration: Burning of waste at high temperature
  - Landfill: Disposal of non-hazardous waste in land
  - Chemical treatment: Neutralization of chemical waste
  - Autoclaving: Steam sterilization of biological waste

## ④ Documentation

- All waste disposal activities are recorded.
- Records are maintained for regulatory compliance.

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# DOCUMENTS

- Documents are written records that provide evidence of activities performed in pharmaceutical manufacturing & quality control.
- They ensure traceability, consistency & compliance with GMP (Good Manufacturing Practices)
- Some common documents include:

1. Batch Formula Record
2. Master Formula Record
3. Standard Operating Procedure
4. Batch Packaging Record
5. Validation Protocols

## OBJECTIVES OF DOCUMENTATION

- To maintain product quality & safety
- To provide proof of compliance with regulatory requirements
- To ensure traceability of every step
- To avoid errors & mix-ups.
- To facilitate investigation & audits

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# Master Formula Record

- MFR is an authorized, approved document that contains complete instructions for manufacturing a specific p'ceutical product.
- It is prepared by R&D Department of the company.
- It serves as a standard reference document for preparing Batch Manufacturing Records (BMR).

## Purpose Of MFR

- To ensure uniformity in production
- To maintain consistent product quality
- To minimize errors & deviations
- To provide a standard guideline for operators
- To comply with GMP requirements.

## Content Of MFR

- Master Formula Record (MFR) is a complete document that provides all instructions & details required for manufacturing a p'ceutical product.
- It includes following details:

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## ① Product Details

This section includes basic identification & description of product:

- Name, logo & address of manufacturing company
- Dosage form
- Brand Name
- Generic Name
- Product Code
- Label claim (Amount of API + Excipients)
- Product description (colour, shape, appearance)
- Batch Size
- Pack size & packaging style
- Shelf life
- Storage conditions (temperature, humidity, light)

## ② Document Identification & Authorization

It includes:

- MFR number & date
- Superseded MFR Number (previous version)
- Effective Batch Number
- Approval by Production & QA head
- It ensures control & traceability

## ③ Flow Chart

- It includes flowchar of manufacturing process.
- It shows steps from dispensing → processing → final product → storage
- This helps in easy understanding of process flow.

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## ④ Equipment Details

- This section contains list of all equipment & machines used
- Equipment identification Number
- Capacity of each equipment
- This ensures correct equipment usage

## ⑤ Special Instructions / Precautions

- It includes safety precautions during manufacturing.
- Special instructions for handling materials.
- This helps in maintaining safety & quality.

## ⑥ Calculations

- It includes calculation of active ingredients.
- Adjustment using LOD/water content for 100% potency.

## ⑦ Manufacturing Process

- It includes step-by-step manufacturing process like:
  - Mixing / blending steps
  - Granulation (if any)
  - Drying condition
  - Compression or filling
  - Environmental conditions (temperature, humidity)
  - This is the core part of MFR

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## ⑧ Packing Process & Yield

- It includes list of packaging materials, quantities & label details.
- MFR also includes expected output of product which helps in checking losses & efficiency.

## Preparation Of MFR

- Preparation of MFR means creating a standard document that contains all instructions required to manufacture a product.
- It is prepared by R&D team or F&D department in collaboration with Production Department.
- It is reviewed & approved by Quality Assurance Department.
- It is prepared by qualified staff like manufacturing or analytical chemists based on experience.

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## Steps To Prepare MFR

- MFR should be devided into Two parts
  - Manufacturing Part
  - Packaging Part
- Now following details are added step by step:

① Product Information
② Manufacturing formula
③ Manufacturing Process
④ Equipment details
⑤ Special instructions
⑥ Calculations
⑦ Quality Checks & Yields
⑧ Packaging instructions

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## RESPONSIBLE DEPARTMENTS

① Primary Responsibility: F & D and Production Department

② Secondary Responsibility: Quality Assurance Department

## ACCOUNTABILITY

- Head - Quality Assurance shall be responsible for implementation of SOP.

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# BATCH MANUFACTURING RECORD

- Batch Manufacturing Record (BMR) is an official written document that contains complete step-by-step instructions & records for manufacturing a specific batch of a product.
- It ensures that product is consistency manufactured & controlled according to approved procedures (MFR).
- It can be knows as BFR

## PURPOSE OF BMR

- To maintain complete traceability of each batch
- To serve as an official & legal document
- To facilicate investigation of errors & complaints
- To provide accountability & responsibility
- To ensure compliance with GMP

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## CONTENTS OF BMR

- BMR contains information from actual manufacturing operations performed based on MFR.
- It contains following details as followed:

### ① PRODUCT INFORMATION

- Name of the Product
- Dosage form (e.g. Tablet, Capsule, liquid)
- Strength of the product
- Batch Number
- Batch Size

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## ② List Of Raw Materials

- Name of each raw materials
- Specified quantity required
- Material identification / code number
- Weighing & dispensing details

## ③ Manufacturing Instructions

- Detailed step-by-step manufacturing procedure
- Equipment to be used
- Processing parameters such as temperature, time & speed

## ④ In-Process Controls

- Tests performed during manufacturing
- Critical parameters such as:
  - Weight variation
  - pH
  - Hardness (for tablets)

## ⑤ Yield Details

- Theoretical Yield
- Actual Yield obtained
- Percentage yield calculation

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## ⑥ Equipment Details

- Name & identification
- Equipment Number
- Equipment cleaning records

## ⑦ Signatures & Approvals

- Signature of the operator
- Signature of the Supervisor
- Approval by the Quality Assurance (QA) Department.

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# STANDARD OPERATING PROCEDURE

A Standard Operating Procedure (SOP) is a written & approved document that provides step-by-step instructions to perform a specific task in a consistent, safe & controlled manner.

## PURPOSE OF SOP

- To ensure uniformity in work
- To maintain product quality
- To reduce errors & variability
- To ensure compliance with GMP/GLP guidelines
- To provide training guidance for employees

## TYPES OF SOP

① Operational SOP: Equipment operation, Safety instructions

② Cleaning SOP: Equipment & area cleaning

③ Quality Control SOP: Testing & Sampling

④ Maintenance SOP: Equipment Servicing

⑤ Safety SOP: Handling hazards & emergencies

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## Contents Of SOP

A well-designed SOP contains the following sections:

### ① Title Page

- Title page includes:
  - Title of the SOP
  - SOP number (unique code)
  - Version or revision number
  - Effective Date

### ② Purpose

- This section explains why the SOP is prepared.
- It describes the objective of the procedure in a clear & concise manner.

### ③ Responsibility

- This section defines roles & duties of personnel; such as:
  - Operator: Performs the tasks
  - Supervisor: Checks & Verifies
  - Quality Assurance (QA): Approves & Ensures compliance

### ④ Materials & Equipment

- This section list all materials, tools & equipments required to perform the task.

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## ⑤ PROCEDURES

- This is the most important part of the SOP.
- It includes:
  - Step-by-step instructions
  - Sequence of operations
  - Process conditions (time, temperature etc.)
- The procedure must be written in clear, simple & logical steps.

## ⑥ SAFETY & PRECAUTIONS

- This section provides safety instructions to protect personnel & equipment.
- It includes information such as:
  - Use of PPE (gloves, masks, goggles)
  - Handling of hazardous materials

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## ⑦ DOCUMENTATION & RECORDS

- This section specifies what records must be maintained.
- Examples: Logbooks, Checklists etc.

## ⑧ DEVIATIONS & CORRECTIVE ACTIONS

- It explains what to do if any error or deviation occurs during the process.
- It also includes corrective & Preventive Actions (CAPA).

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## ⑨ Approval Section

- This section ensures that the SOP is official & authorized.
- It includes information such as:
  - Prepared by
  - Reviewed by
  - Approved by

## Characteristics Of A Good SOP

- Clear & easy to understand
- Step-by-step format
- Written in simple language
- Approved by QA
- Regularly reviewed & updated.

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# Quality Audit

- Quality Audit is a systematic, independent & documented examination of activities, processes & records to verify whether they comply with approved procedures (SOPs), GMP guidelines & regulatory requirements.
- It is a process of checking whether work is being done correctly & according to standards.

## Objectives Of Quality Audit

- To ensure compliance with GMP & Regulatory guidelines.
- To verify that SOPs & Procedures are properly followed.
- To identify errors, deviations & non-compliance
- To improve quality systems & processes
- To ensure product safety, quality & consistency

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## Types Of Quality Audit

It can be classified into 3 types:

1. Internal Audit
2. External Audit
3. Regulatory Audit

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## ① Internal Audit

- It is conducted by the company itself.
- It helps in regular monitoring & improvement.

## ② External Audit

- It is conducted by clients, suppliers or third parties.
- It ensures business & quality requirements are met.

## ③ Regulatory Audit

- It is conducted by government authorities.
- Example: Inspections by FDA, CDSCO
- It ensures compliance with legal standards.

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## Scope Of Quality Audit

- A quality audit may cover the following areas:

- Production / Manufacturing
- Quality Control (QC) laboratory
- Quality Assurance (QA) System
- Documentation (SOPs, BMR, Records)
- Equipment & Maintenance
- Personnel training & hygiene
- Storage & Distribution

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## Steps Involved In Quality Audit

- Planning: Preparation of Audit Schedule & Checklist.
- Execution: Actual inspection of processes, records & facilities.
- Observation: Identification of deviation, errors, & non-compliance.
- Reporting: Preparation Audit Repost.
- CAPA: Actions taken to correct & prevent future issues.

## Advantages Of Quality Audit

- Improves overall quality system
- Reduces risk of errors & defects
- Ensures regulatory compliance
- Enhances process effeciency
- Supports continuous improvement

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# QUALITY REVIEW

- Quality Review is a periodic evaluation of processes, products & records to ensure that they meet quality standards & regulatory requirements (GMP).
- It simply means regularly checking whether everything is working properly & producing quality products.

## OBJECTIVES OF QUALITY REVIEW

- To ensure consistent product quality
- To review manufacturing & control processes
- To identify trends, deviations & problems
- To improve process performance
- To ensure compliance with GMP

## ELEMENTS OF QUALITY REVIEW

- Review of batch records (BMR / BFR)
- Review of deviations & CAPA
- Review of complaints & Recalls
- Review of stability data
- Review of analytical results (QC data)
- Review of yield & process consistency.

## IMPORTANCE OF QUALITY REVIEW

- Helps in continuous improvement
- Detects repeated problems (trends)
- Ensures product safety & quality
- Supports regulatory inspections.

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# QUALITY DOCUMENTATION

- Quality Documentation refers to all written records & documents that describe & prove that activities are performed according to GMP & approved procedures.

## OBJECTIVES OF QD

- To ensure compliance with GMP
- To provide evidence of performed activities
- To maintain consistency & standardization
- To ensure traceability of all operations
- To support audits & inspections.

## TYPES OF QD

Quality documentation mainly includes two categories:

① Documents
② Reports

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## ① DOCUMENTS

- Documents are written & approved instructions that describe how an activity should be performed.
- They act as guidelines or instructions:
- Example: - SOP (Standard Operating Procedure)
  - MFR (Master Formula Record)
  - Specifications
  - Protocols

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## ② REPORTS

- Reports are written records that show the results, findings or outcomes of activities performed according to documents.
- They act as proof or evidence.
- Examples :
  - Batch Manufacturing Record (BMR)
  - Validation reports
  - Audit reports
  - Investigation reports
  - Test / Analysis reports

## IMPORTANCE FOR QD

- Ensure regulatory compliance (GMP)
- Provides evidence during audits
- Helps in investigation of errors
- Maintains product history & traceability

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# DISTRIBUTION RECORDS

- Distribution records are documents that keep complete details about how & where a p'ceutical product is distributed after manufacturing.
- It is generally maintained by Warehousing Department.

## DETAILS IN DISTRIBUTION RECORDS

Distributions records usually include:

- Name, strength & dosage form of product
- Batch / Lot Number
- Total quantity distributed
- Date of distribution
- Name & address of Recipient (Wholesaler, distributor or pharmacy)
- Transport details
- Storage conditions

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## IMPORTANCE OF DISTRIBUTION RECORDS

- To track movement of products from manufacturer to market.
- To help in product recall if any problem occurs
- To ensure transparency & accountability
- To maintain regulatory compliance.

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